

PATUXENT RIVER NAS, MD, USA

WMO: 724040

Lat: 38.300N

Lon: 76.417W

Elev: 12

StdP: 101.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.3	(c) -6.3	(d) -18.2	(e) 0.8	(f) -6.5	(g) -15.7	(h) 1.0	(i) -4.4	(j) 12.9	(k) -1.4	(l) 11.9	(m) -1.2	(n) 5.0	(o) 330	(p) 0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	33.7	24.7	32.3	24.3	31.0	23.8	26.3	31.2	25.5	30.2	24.9	29.3	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.1	28.8	24.1	19.0	28.0	23.5	18.4	27.5	81.6	31.3	78.3	30.2	75.5	29.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.2	8.2	-11.9	36.4	2.8	1.6	-13.9	37.6	-15.6	38.6	-17.1	39.5	-19.2	40.7
			-12.9	28.1	2.6	1.2	-14.8	29.0	-16.3	29.7	-17.8	30.4	-19.7	31.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	2.4	3.8	7.6	13.6	18.9	23.7	25.9	24.9	21.6	15.7	9.5	5.5		
	DBStd	9.29	5.66	5.16	5.11	4.72	4.21	3.27	2.69	2.61	3.32	4.35	4.42	4.97		
	HDD10.0	782	246	184	110	16	0	0	0	0	0	4	62	159		
	HDD18.3	2226	496	408	336	157	46	3	0	0	9	106	267	399		
	CDD10.0	2413	9	11	34	123	276	411	492	463	349	181	46	18		
	CDD18.3	815	0	0	2	14	63	163	234	204	108	24	2	1		
	CDH23.3	6816	2	1	14	107	449	1348	2279	1795	691	127	3	1		
	CDH26.7	2249	0	0	2	26	126	450	876	581	166	23	0	0		
(14) Wind	WSAvg	3.8	4.2	4.2	4.4	4.3	3.7	3.4	3.1	3.0	3.4	3.6	3.9	3.8		
(15) Precipitation	PrecAvg	1034	85	78	99	73	103	93	94	96	89	77	72	83		
	PrecMax	1544	175	178	258	209	286	188	266	239	203	168	152	181		
	PrecMin	718	11	21	20	11	12	23	21	16	2	0	8	12		
	PrecStd	235	38	39	57	40	71	50	62	61	56	43	41	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	21.2	25.3	29.7	32.4	34.6	36.7	34.8	33.0	29.0	23.0	21.3		
		MCWB	15.5	15.2	15.9	18.4	23.2	24.1	25.4	25.3	23.9	22.4	17.9	16.8		
	2%	DB	17.0	17.5	21.3	26.3	29.8	32.5	34.1	32.7	30.2	26.4	20.7	18.5		
		MCWB	13.7	12.9	14.8	18.1	22.1	23.8	25.0	24.8	23.1	20.8	16.7	15.2		
	5%	DB	13.9	14.1	18.5	23.7	27.7	31.0	32.4	31.2	28.5	24.4	18.6	16.0		
		MCWB	10.9	10.1	13.3	17.2	21.2	23.3	24.6	24.3	22.4	20.0	15.3	13.3		
	10%	DB	11.0	11.4	16.0	21.4	26.0	29.2	31.0	29.8	27.1	22.5	16.7	13.0		
		MCWB	8.1	7.9	11.4	15.9	20.2	22.7	24.2	23.6	21.8	18.4	13.4	10.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	17.0	18.0	21.3	24.4	26.2	27.7	27.3	25.8	23.9	19.4	18.5		
		MCDB	18.8	20.0	23.1	25.6	30.3	31.3	32.0	31.1	29.9	26.7	21.7	21.0		
	2%	WB	14.1	14.0	16.0	19.6	23.1	25.2	26.6	26.0	24.6	22.3	17.8	15.7		
		MCDB	16.6	16.9	20.2	24.1	27.8	29.9	31.3	30.9	28.1	25.7	19.8	17.9		
	5%	WB	11.5	11.0	14.1	18.1	22.1	24.6	25.7	25.3	23.9	20.8	15.7	13.6		
		MCDB	13.6	13.5	17.7	22.3	26.6	29.0	30.6	29.7	27.1	23.6	18.0	15.7		
	10%	WB	8.2	8.3	11.9	16.8	21.0	23.9	25.0	24.6	22.9	19.1	14.0	10.7		
		MCDB	10.3	11.0	15.1	20.9	25.0	28.0	29.8	28.6	26.0	21.8	16.2	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	9.1	9.8	9.1	8.8	8.6	8.3	8.4	9.1	9.0	8.4		
		MCDBR	12.0	12.7	13.2	12.7	11.1	10.5	10.2	9.4	9.7	10.4	11.1	11.3		
	5% WB	MCWBR	9.8	9.7	8.5	6.9	5.7	4.6	4.2	4.1	4.7	6.1	7.7	9.1		
		MCWBR	11.6	11.8	11.9	11.1	9.9	9.2	8.9	8.3	7.9	8.9	9.2	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.319	0.354	0.393	0.434	0.485	0.516	0.507	0.410	0.360	0.326	0.311		
	taud		2.511	2.479	2.426	2.332	2.250	2.157	2.084	2.093	2.341	2.485	2.539	2.546		
	Ebn at Noon		874	911	907	885	851	806	776	772	836	853	849	847		
	Edh at Noon		78	91	106	124	137	150	161	156	115	90	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.20	3.02	4.04	5.09	5.80	6.37	6.18	5.53	4.48	3.41	2.49	1.94		
	RadStd		0.17	0.32	0.34	0.36	0.64	0.47	0.30	0.30	0.45	0.37	0.21	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+87

Nomenclature: See separate page